Unstable, Inverted and Re-Ranked: What Natural Zero-Day Traffic Reveals About Learned Network Intrusion Detectors

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***Abstract*—Operators deploy learned network intrusion detection systems (NIDS) to catch attacks absent from their training data, yet zero-day performance is usually measured once, on one protocol, with scores borrowed from multi-class image recognition. We evaluate 12 detectors from five families, over five seeds, under both class hold-out (LOACO) and natural temporal novelty, in which attack classes first appear after the training period, in a pre-registered, leakage-controlled benchmark on four NetFlow-v3 datasets and LYCOS-IDS2017. We report three findings. First, natural novelty raises the seed variability of every supervised score 1.7–17-fold over LOACO; on ToN-IoT a nearest-neighbour detector's AUC varies by 0.33–0.40 between seeds, so the same model can rank first or last. Second, max-softmax and energy scores are structurally invalid in binary detectors: a confidently detected novel attack reads as normal, so MSP scores 0.163 where the attack probability scores 0.999 on the same models. Third, detector rankings flip between protocols, and a plain XGBoost classifier matches or beats every dedicated novelty family in four of seven evaluations, contrary to prior comparisons. We report four failed pre-registered hypotheses, derive an assurance checklist for operators, and release all splits and code.**

***Index Terms*—network security management, intrusion detection, zero-day attacks, service assurance, evaluation methodology, NetFlow**

I. Introduction

Anetwork operator who deploys a learned network intrusion detection system (NIDS) is buying protection against attacks that are not in its training data. Before trusting the detector, the operator needs evidence that it raises an alarm when a new attack class first appears, and that this behaviour survives the routine retraining of an operational service. This is a security-management decision: a detector that misses novel attacks creates false assurance, and one whose zero-day coverage changes from one retraining to the next cannot be operated predictably.

The problem. The evidence operators receive is usually a single training run evaluated by leave-one-attack-class-out (LOACO): one attack class is removed from a randomly split benchmark and the detector must flag it at test time [22], [23], [20]. Novelty is often scored with confidence scores designed for multi-class image recognition, such as maximum softmax probability (MSP) [37] and energy [38]. In deployment, however, a new class arrives later in time, together with drifting benign traffic, and the detector is retrained repeatedly. Security-ML research has shown that evaluation shortcuts inflate results [1], [2], [3], [4], and recent work has begun to study time-aware NIDS splits [25], [15], [16].

The gaps. Three questions that matter to an operator remain open. (G-1) *Stability:* prior work reports at most aggregate run-to-run variance [14]; whether a detector's zero-day result is stable across retraining, per attack class and under realistic novelty, is unknown. (G-2) *Score validity:* MSP-type scores have been observed to perform poorly, even below chance, on unseen attacks in multi-class NIDS [13], [14], but no work explains why or establishes whether they are valid at all for the common two-class benign-versus-attack detector. (G-3) *Rankings:* existing studies apply class hold-out and temporal splits to a single proposed method [18], [19] or rank supervised classifiers only [15]; whether the ranking of detector *families* (classifier-confidence, representation-distance, benign-only, contrastive and supervised) survives the move from LOACO to natural novelty is unknown, and prior comparisons disagree on whether supervised models or novelty detectors handle unknown attacks better [21], [17].

Research questions. **RQ1** How stable are zero-day results across training seeds, per class, under LOACO and under natural temporal novelty? **RQ2** Which novelty scores are valid for binary NIDS, and why do confidence scores fail? **RQ3** Do detector-family rankings agree between the two protocols, and do dedicated novelty detectors beat a plain supervised classifier?

Approach. We build a pre-registered, leakage-controlled benchmark on five public datasets: four NetFlow-v3 datasets [31] and LYCOS-IDS2017 [8]. Flows are grouped into time blocks before windowing, windows are deduplicated by hash, and every split is checked fail-closed. On day-scheduled captures, where attack classes occupy separate days [16], the time-ordered split leaves classes that appear only in the test period; we use them as a natural zero-day protocol alongside LOACO. We evaluate 12 detectors in five families, each over five seeds, from one code commit.

Contributions.

- **C1 Hidden instability (RQ1).** Natural novelty raises the per-class seed standard deviation of every supervised score 1.7–17× over LOACO; on ToN-IoT, kNN varies by 0.33–0.40 AUC between seeds on four of five classes. Seeded benign-only detectors stay stable. A

single LOACO run therefore hides how unpredictable a supervised detector is on real novelty.

- **C2 Why confidence scores invert (RQ2).** We identify the mechanism by which MSP and energy fail in binary NIDS: a confidently detected novel attack has high softmax confidence and reads as in-distribution. MSP scores 0.163 on LOACO D2 where P(attack) scores 0.999 on the same models, and MSP is below chance even in-distribution (0.075–0.394). The classifier's own attack probability does not invert on average.
- **C3 Rankings flip between protocols (RQ3).** Across five families, the leader changes with protocol and dataset: on ToN-IoT the benign-only lead over representation scores grows from $+0.022$ under LOACO to $+0.376$ under natural novelty, while representation scores fall to chance.
- **C4 A supervised reference is hard to beat (RQ3).** Contrary to comparisons favouring unsupervised detectors [21], [17], XGBoost matches or beats the best novelty family in four of seven evaluations, and the contrastive zero-day detector CLAD [22] never leads.
- **C5 Pre-registration and release.** To our knowledge this is the first pre-registered NIDS benchmark; we report its four failed hypotheses and release splits, code and amendments.

From these findings we derive a six-point checklist for assuring a detection service before deployment (Section VII). Section II gives background, Section III related work, Section IV the benchmark, Section V the experiments, Section VI the results and Section VII the discussion.

II. Background

A. Flow-based NIDS and windowing

Flow-based NIDS summarize each connection as a record of counters and timings exported by network monitoring infrastructure. Sarhan et al. converted several public captures into a common NetFlow feature set so that detectors can be compared across networks [32]; the third version (v3) adds temporal features and per-flow start and end timestamps [31], which make strictly time-ordered splits possible. Sequence detectors consume windows of T consecutive flows. Windowing creates two leakage risks: overlapping windows can place near-identical samples in training and test, and flows from one attack episode can straddle the split. Grouping flows into contiguous time blocks before windowing and deduplicating windows by hash remove both.

B. Zero-day evaluation protocols

Three protocols are compared in this paper (Fig. 1, stage 4). *LOACO* removes one attack class from the training pool; the test set holds benign windows plus that class, and splits are random, here grouped by 1-hour blocks. *Natural temporal novelty* splits flows in time; attack classes that occur only after the training period are unseen by construction, and benign traffic drifts as it would in operation. *Purged block LOACO*, for short captures, assigns short time blocks to splits within each day and drops blocks adjacent to test or validation blocks, following purged cross-validation for time series [45]. LOACO asks whether a detector generalizes to a missing class under the training distribution; natural novelty asks the operator's question, under real temporal shift.

C. Novelty scores and the binary pitfall

Out-of-distribution (OOD) detection assigns each input a score that should be higher for inputs unlike the training data. *Classifier-confidence* scores read the output layer: MSP scores $1 - \max_k p_k$ [37] and energy scores $-\log \sum_k e^{f_k}$ over logits f_k [38]. *Representation-distance* scores measure how far hidden features lie from benign training features, e.g., Mahalanobis [39] or nearest-neighbour distance [40]. *Benign-only* detectors never see attacks [41], [42]. *Contrastive* detectors such as CLAD [22] learn an embedding from benign and known attacks. MSP and energy measure how sure a classifier is, not which class it chose. In a two-class NIDS, a new attack that is confidently classified as *attack* receives a low MSP score and therefore reads as in-distribution: the expected outcome of successful detection becomes a missed novelty. In a binary detector this is not an edge case but the expected outcome whenever a novel attack is detected.

III. Related Work

Evaluation pitfalls and dataset quality. Sommer and Paxson argued that anomaly detection outside a closed world is harder than benchmark scores suggest [1]. Arp et al. catalogued ten pitfalls in security ML, including sampling bias and data snooping [2], and TESSERACT showed that ignoring time inflates malware-detection results [3]. For NIDS, Apruzzese et al. found that evaluations rarely reflect operational conditions [4], and Husák et al. reported, from a network-management perspective, that ML intrusion detection research favours ML metrics over operational outcomes [5]. Dataset audits found labelling and flow-construction errors in CIC-IDS2017/2018 [6], [7], corrected in LYCOS-IDS2017 [8]; design smells in benchmark NIDS datasets that cause severe experimental bias [9]; and metadata features that let models shortcut the task [10]. These works diagnose problems; we measure how one of them, the choice of zero-day protocol, changes which detector an operator would deploy.

Zero-day and open-set NIDS. A recent survey in IEEE TNSM concludes that ML intrusion detectors detect known attacks well but unknown, atypical and polymorphic attacks poorly [11]. Sarhan et al. framed zero-day detection as zero-shot learning and linked low detection rates to large Wasserstein distance from known attacks [20]. Zoppi et al. compared 47 algorithms on unknown attacks and found unsupervised meta-learners strongest [21]. Wilkie et al. proposed CLAD, a contrastive loss trained on benign and known attacks [22]; Jodayree et al. combined a transformer with energy scoring and extreme-value calibration, reporting an AUROC of 0.975 under leave-one-attack-out on IoMT traffic [23]; Thi and Gurusamy detect open-set attacks with a model trained on benign traffic only [24]; and Baye et al. benchmarked deep open-set recognition for NIDS at NOMS [12]. Softmax-confidence

TABLE I
POSITIONING AGAINST REPRESENTATIVE PRIOR WORK (–: NOT ASSESSED).

Work	Zero-day protocol	X-data	Families	Seed var.	Pre-reg.
Sarhan et al. [20]	zero-shot hold-out	no	supervised	–	no
Zoppi et al. [21]	unknown attacks	no	sup./unsup./meta	–	no
Layeghy & Portmann [29]	–	yes	supervised	–	no
Wilkie et al. [22]	LOACO	no	contrastive, anomaly	–	no
Jodayree et al. [23]	leave-one-out	1 pair	energy + EVT	5 seeds	no
This work	**LOACO + natural + purged**	**9 pairs**	**5 (12 det.)**	**per class**	**yes**

TABLE II
DATASETS. WINDOWS ARE COUNTED AFTER GLOBAL DEDUPLICATION.

ID	Dataset	Flows	Windows	Att. cl.	Role
D1	NF-CSE-CIC-IDS2018-v3	20,115,529	1,255,180	14	LOACO, NN, transfer
D2	NF-UNSW-NB15-v3	2,365,424	147,801	9	LOACO, transfer
D3	NF-ToN-IoT-v3	27,520,260	1,553,191	9	LOACO, NN, transfer
D4	NF-BoT-IoT-v3	16,933,808	890,852	4	transfer target
D5	LYCOS-IDS2017	1,837,498	114,783	13	NN, purged LOACO

baselines have scored below chance on held-out attacks in multi-class NIDS [13], and softmax entropy and energy trailed uncertainty-aware models [14]; these observations were made for multi-class detectors and without an explanation tied to the binary setting. All of these works evaluate under class hold-out or in-distribution.

Drift and temporal evaluation of NIDS. INSOMNIA applied time-aware evaluation to intrusion detection and found that accounting for drift is vital [25]; ReCDA [26] and continual-learning NIDS [27] adapt detectors to drifting traffic. Recent work examines temporal protocols directly: Abdessamad and Tazi found that temporal splits on ToN-IoT leave unseen attack classes in test and that classifier rankings depend on the protocol [15]; Moczkodan and Ragab found a chronological CIC-IDS2017 split degenerate because attacks were scheduled by day [16]; Mesca et al. compared classifiers under random, attack-held-out and leave-one-out protocols on industrial traffic [17]; and ThreatFormer-IDS [18] and Alhalangy and Eltayeb [19] evaluate a single proposed detector under both chronological and held-out-family splits. None compares detector families under class hold-out versus natural temporal novelty, the question an operator faces when choosing a detector.

Cross-domain generalization. NetFlow NIDS are commonly evaluated in-distribution, e.g., the graph-based E-GraphSAGE [28]. Layeghy and Portmann [29] and Cantone et al. [30] found that near-perfect within-dataset performance falls towards chance across datasets; These measure transfer between networks but not novelty within one network.

Positioning. Table I summarises the gap: prior work evaluates either class hold-out or cross-dataset transfer, typically for one detector family, without per-class seed variability or pre-registration.

IV. BENCHMARK METHODOLOGY

Fig. 1 shows the benchmark as four stages: the input datasets, a leakage-controlled processing pipeline, the backbone and its 12 detectors, and the evaluation protocols that produce the outputs C1–C6. Data, splits, backbone and detectors were fixed in advance, and every protocol change was recorded as a dated, hashed amendment before the runs it governs (Section IV-E).

A. Stage 1: Input datasets

All five datasets are public, were downloaded from their official sources and are recorded in a manifest with SHA-256, row count and column list (Table II). D1–D4 are the NetFlow-v3 releases of the University of Queensland [31], which re-extract each capture's packet traces into 53 NetFlow features with per-flow start and end timestamps in milliseconds, using a common feature set across datasets [32].

- **D1 CSE-CIC-IDS2018** [33] was generated by the Canadian Institute for Cybersecurity on a cloud-hosted enterprise testbed with scripted benign users and attack machines; our copy spans 14 Feb–2 Mar 2018. Its 14 attack labels (brute force, botnet, DoS, DDoS, web attacks, infiltration) map onto six families. Labelling and feature-generation errors in the original release have been documented [7].
- **D2 UNSW-NB15** [34] was created in the UNSW Canberra Cyber Range with a traffic generator that mixes real normal activity with nine synthetic attack families (e.g., exploits, fuzzers, reconnaissance, backdoors); it spans 22 Jan–18 Feb 2015.
- **D3 ToN-IoT** [35] comes from an IoT/IIoT testbed with edge, fog and cloud layers and nine attack types (scanning, DoS, DDoS, ransomware, backdoor, injection, cross-site scripting, password and man-in-the-middle); it spans 23–29 Apr 2019.
- **D4 BoT-IoT** [36] simulates IoT devices under botnet attacks (DDoS, DoS, reconnaissance and data theft); it spans May–Jun 2018 but contains only 1,315 benign windows, so it serves only as a transfer target.
- **D5 LYCOS-IDS2017** [8] re-processes the CIC-IDS2017 packet traces (five working days, 3–7 Jul 2017) with a corrected flow extractor and relabelling that fix documented flow-construction and labelling errors [6], [7]. We ran the authors' labelling script unmodified, kept its 73 own features and do not use D5 for transfer.

B. Stage 2: Leakage-controlled processing

Each step addresses a known source of optimistic bias [2], [9], [10].

1) *Audit and drop.* We record rows, types, missing values and class counts, then drop identifier columns (addresses, ports, DNS query ID), whose values identify hosts rather than behaviour [10], and minimum/maximum TTL, a known label-leakage artifact. Timestamps are kept only for ordering and splitting.
2) *Clean with training statistics only.* Infinite values become missing and are imputed with the training median; non-negative heavy-tailed counts are transformed with

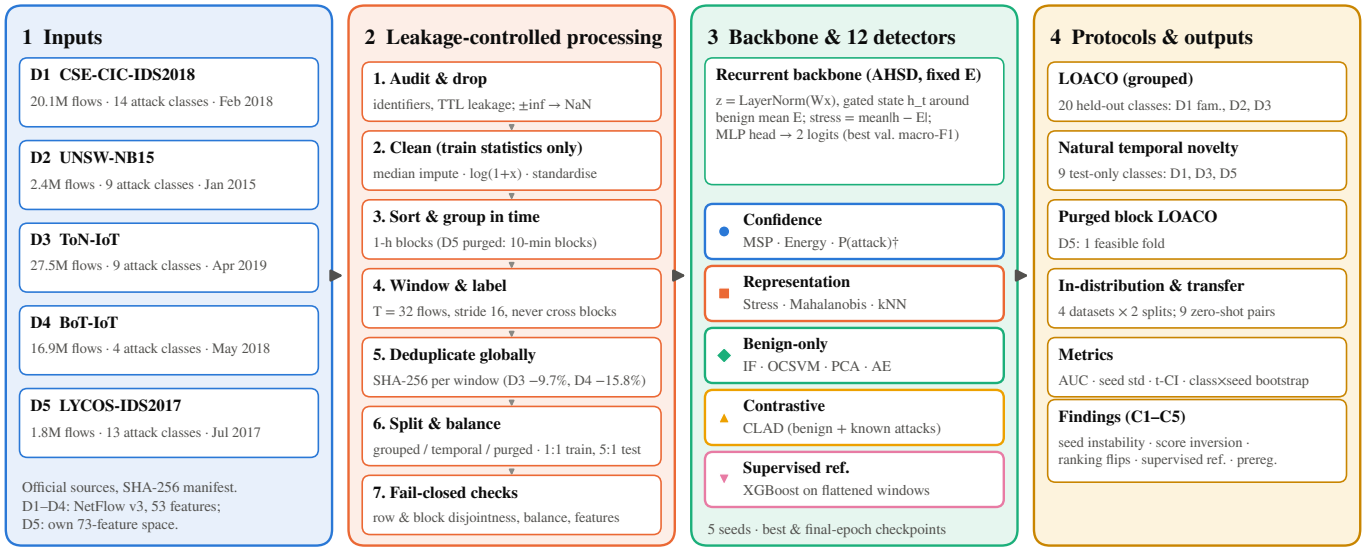


Fig. 1. Benchmark methodology. (1) Five public datasets enter with checksummed manifests. (2) Seven processing steps turn flows into deduplicated, time-blocked windows and fail-closed checked splits. (3) One recurrent backbone supplies the classifier-confidence and representation-distance scores; benign-only, contrastive and XGBoost detectors are trained on the same splits. (4) Five evaluation settings, each over five seeds, produce the six results C1–C6.

$\log(1+x)$; features are standardized with training mean and deviation; zero-variance features and one of each pair with $|r| \geq 0.999999$ are removed. For D1–D4 the intersection leaves 41 common features. No statistic is ever fitted on test or transfer-target data.

3) *Sort and group in time.* Flows are sorted by start time and grouped into contiguous 1-hour blocks (10-minute blocks for D5's purged split), so that flows of one attack episode stay together.
4) *Window and label.* Windows of T=32 consecutive flows with stride 16 are cut inside blocks and never cross a block boundary. A window is an attack if any flow is an attack, and its class is the majority attack class.
5) *Deduplicate globally.* Every window is hashed over its raw features and exact duplicates are removed across the whole dataset, so that no window appears in both training and test (9.7% of D3 and 15.8% of D4 windows were duplicates).
6) *Split and balance.* Blocks, never individual flows or windows, are assigned to splits (Section IV-D). Training sets are balanced 1:1 (7,000 benign and 7,000 attack windows); validation and test sets use a 5:1 benign-to-attack ratio, capped at 2,500 benign and 500 attack windows.
7) *Fail-closed checks.* The pipeline stops if any check fails: row and block disjointness across splits, deduplication, balance, class composition and feature lists. 538 checks ran at preprocessing for D1–D4 and further checks for every derived split (e.g., 32 for D5's purged split); all passed.

C. Stage 3: Backbone and detectors

Backbone. All supervised scores are read from one recurrent backbone, the fixed-equilibrium variant of a homeostatic sequence detector. Each flow vector is embedded as $z_t = \mathrm{LayerNorm}(Wx_t) \in \mathbb{R}^{128}$. A state h_t, initialised at a benign reference E, tracks the input through per-step, per-channel gates:

$$h_t = h_{t-1} + \alpha_t \odot (z_t - E) - \beta_t \odot (h_{t-1} - E),$$
$$[\alpha_t, \beta_t] = \sigma\big(W_g[z_t, h_{t-1}]\big).$$

E is not learned: it is the mean embedding of benign training windows, tracked by an exponential moving average during training and set exactly afterwards, and it stays fixed within a window. Homeostatic *stress* is $\frac{1}{TD}\sum_{t,d}|h_{t,d} - E_d|$. A two-layer MLP head (linear–GELU–linear) reads $[h_T,\ \frac{1}{T}\sum_t h_t,\ \text{stress}]$ and outputs two logits; its hidden layer is the penultimate representation. We chose this backbone because it exposes a representation-distance score inside the model.

Detectors. Twelve detectors in five families score each test window; higher scores always mean more anomalous, and orientation was fixed before any run.

- *Classifier-confidence* (from the backbone's logits): MSP, energy, and P(attack), the softmax attack probability, which was added after the last confirmatory gate and is marked †.
- *Representation-distance* (from the backbone's states): stress; Mahalanobis distance to a Ledoit–Wolf Gaussian fitted on benign penultimate features [39]; and distance to the 10th benign nearest neighbour of ℓ_2-normalised penultimate features [40].
- *Benign-only* (fitted on benign training windows, never seeing attacks): Isolation Forest with 200 trees [41],

one-class SVM with RBF kernel and ν=0.1 [42], PCA reconstruction error at 95% retained variance, and a fully connected autoencoder.
- *Contrastive*: CLAD [22], trained on benign and known attacks and implemented from its paper, with the details it leaves open fixed before training.
- *Supervised reference*: XGBoost [43] on flattened windows (300 trees, depth 6, learning rate 0.1), scored by its attack probability.

D. Stage 4: Evaluation protocols

grouped_random shuffles blocks into 60/20/20 train/validation/test. *temporal_gap* orders blocks in time, takes the first 60% for training, discards one gap block, and uses the next 20% for validation and the last 20% for test. On these splits we run five settings.

- *LOACO* (grouped_random): a class is held out if it has at least 200 training-pool windows (by majority label) and at least one test window; training and validation exclude every window containing any flow of that class. Every resulting fold has at least 81 held-out test windows.
- *Natural temporal novelty* (temporal_gap): classes that occur in the test period but not in the training period are scored as unseen against test benign windows. Validation is the last 20% of the training period (train-tail validation), because on D1 every temporal validation window contains a natural-novelty flow.
- *Purged block LOACO* (D5): 10-minute blocks are split 60/20/20 within each capture day; training and validation blocks adjacent to a test block, and training blocks adjacent to a validation block, are removed [45]. Here a held-out class also needs at least 50 test windows.
- *In-distribution*: binary detection on the test split of both split schemes.
- *Zero-shot transfer*: models trained on D1–D3 are applied to every other NetFlow-v3 dataset, including D4; every detector reference (E, benign-only fits, Mahalanobis, kNN and CLAD references) comes from the source network, and nothing is fitted on the target.

E. Pre-registration and provenance

Research questions, metrics and acceptance gates were written to a file and hashed before each experiment; every result file records that hash, the code commit and the dataset hashes. Six dated, hashed amendments changed the plan, each before the runs it governs: two because data counts made a frozen rule infeasible (an empty validation split; zero LOACO folds on D5), three that set a new direction after a gate returned STOP, and one that added P(attack) and the family split after the last gate. Four confirmatory gates were evaluated exactly as frozen. The final benchmark reported here is descriptive: it applies no thresholds and reports intervals rather than significance tests.

V. Experimental Setup

Training. The backbone is trained with AdamW (learning rate 10^{-3}, weight decay 10^{-4}), cosine annealing, gradient clipping at 1.0, batch size 256 and 15 epochs, with no per-detector tuning. The checkpoint with the best validation macro-F1 is primary (for CLAD, the best validation AUROC). The final-epoch state of the same training run is also kept and scored as a sensitivity analysis. XGBoost uses subsampling of 1, which makes it deterministic.

TABLE III
Protocol feasibility: feasible LOACO folds per split and natural-novelty (NN) classes.

Dataset	LOACO, grouped	LOACO, temporal	NN classes
D1	6 (fam.)	0	2
D2	8	8	0
D3	6	0	5
D5	0	0	2
D5 purged block	1 (dos_hulk)		–

Evaluations. LOACO covers D1 families, D2 and D3 (6, 8 and 6 folds); natural novelty covers D1, D3 and D5 (2, 5 and 2 classes); purged LOACO covers D5 (1 fold); in-distribution detection covers D1, D2, D3 and D5 under both splits; and transfer covers 9 source–target pairs.

Metrics and statistics. Seeds are 17, 23, 42, 101 and 202, and a seeding regression test runs before every experiment. For each detector and class we report ROC AUC as the mean over seeds with a Student-t 95% interval (n=5, t=2.776), the seed standard deviation and the fraction of seeds below 0.5. Family means average the family's detectors and the evaluation's classes, with a 95% bootstrap interval that resamples classes and seeds independently (2,000 resamples). In-distribution results additionally report PR-AUC, macro-F1, MCC and TPR at 1% FPR.

Compute and provenance. The 160 final runs used one code commit and a CPU-only cloud container (4 vCPUs, Intel Xeon at 2.10 GHz, no GPU); together they took 4.27 h of wall-clock time (96 s per run on average). The report scripts refuse to run if code commits differ within an experiment. One run interrupted by a container restart was resumed from the same commit.

VI. Results

All values are mean AUC over five seeds; brackets give 95% intervals. We first state the evaluation setting, then the four findings C1–C4, then results that confirm prior work, and finally the pre-registered gates (C5).

A. Evaluation setting: where each protocol is available

As observed for other day-scheduled captures [16], [15], a time-ordered split puts every attack class of D1 and D3 wholly in the training or wholly in the test period (D1: 12 and 2 of 14 classes; D3: 4 and 5 of 9). Temporal LOACO therefore has no feasible fold on D1 and D3, while natural novelty offers 2–5 classes on D1, D3 and D5 (Table III); D2 supports LOACO under both splits but has no test-only class. On D5 the purged 10-minute split kept 125 of 219 blocks and yielded one LOACO fold (dos_hulk). Natural novelty is

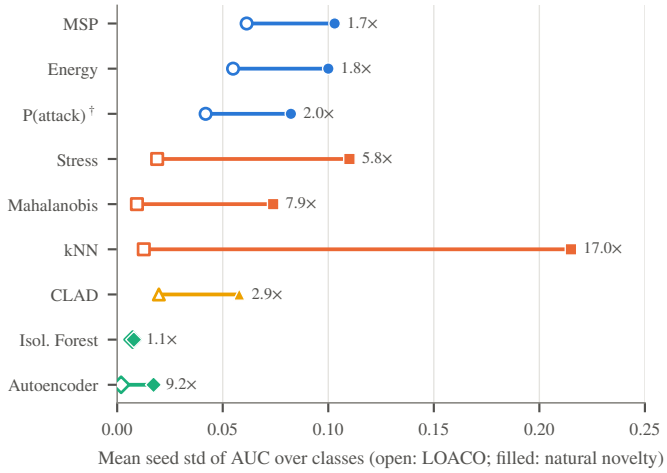


Fig. 2. Seed variability of AUC per detector: open marker LOACO, filled marker natural novelty, labelled with the ratio. Every supervised score becomes 1.7–17× less stable under natural novelty; the seeded benign-only detectors stay near zero. OCSVM, PCA and XGBoost are deterministic and omitted.

thus often the only zero-day protocol available on time-ordered captures.

B. C1: Natural novelty exposes instability that LOACO hides

Every supervised score is less stable under natural novelty than under LOACO (Fig. 2), by 1.7× (MSP) to 17× (kNN). Representation-distance scores are the most affected: their mean seed standard deviation rises from at most 0.019 under LOACO to 0.074–0.215. The two seeded benign-only detectors stay below 0.018, below every supervised score's minimum of 0.074; the autoencoder's large ratio comes from a near-zero LOACO baseline and one class (D5 portscan). On D3 natural novelty, kNN has a seed standard deviation of 0.33–0.40 AUC on four of five classes, so the same detector can rank first or last depending on the seed alone. **Answer to RQ1:** under realistic novelty, a supervised detector's zero-day result depends strongly on the training run, and LOACO understates this by up to 17×.

C. C2: Why confidence-magnitude scores invert in binary NIDS

MSP and energy fall below 0.5 in all four evaluations where the backbone separates unseen attacks well; the attack probability does not (Fig. 3). The mechanism is visible on LOACO D2: MSP is below 0.5 on every seed of all eight classes, while P(attack) reaches 0.997–1.000 on the same models. The classifier detects the held-out attacks with high confidence, and MSP, which measures confidence rather than the predicted class, reads that confidence as normality. The same inversion appears in-distribution, where MSP ranges from 0.075 (D3) to 0.394 (D1) under grouped splits on D1–D3 while P(attack) is 0.812–0.999. Prior multi-class observations of below-chance MSP [13] are consistent with this mechanism. P(attack) falls below 0.5 only on individual classes (D3 LOACO ddos 0.35 and scanning 0.46; D5 natural-novelty portscan 0.16), and

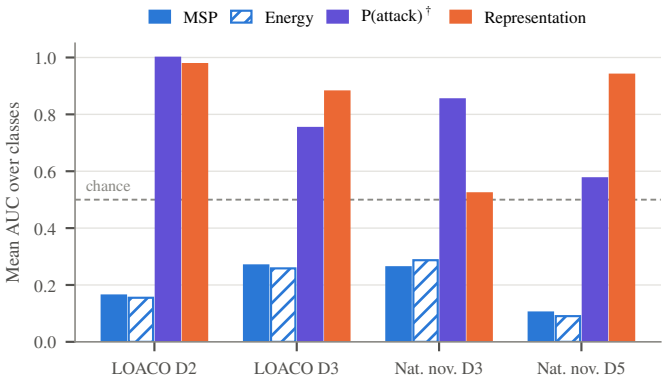


Fig. 3. MSP and energy fall below chance wherever the classifier detects unseen attacks confidently, while the attack probability P(attack) and representation distance stay above chance in the mean.

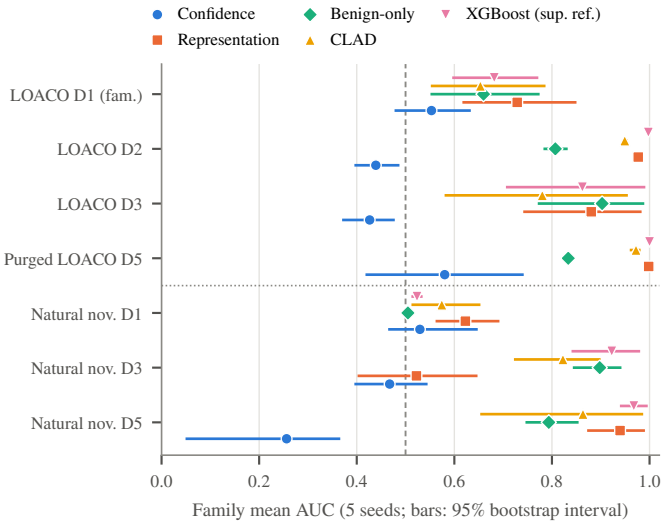


Fig. 4. Family mean AUC per evaluation (markers) with 95% bootstrap intervals over classes and seeds (bars). The leading novelty family changes between LOACO (top four rows) and natural novelty (bottom three); XGBoost (triangles down) is the supervised reference. Dashed line: chance.

representation distance is valid in the mean, although on natural novelty D3 its interval includes 0.5. **Answer to RQ2:** max-softmax and energy are invalid novelty alarms on binary detectors; the attack probability or a representation distance should be used instead.

D. C3: Detector-family rankings flip between protocols

The leading novelty family changes across evaluations (Fig. 4; per-class detail in Fig. 5): representation-distance scores lead in five evaluations and benign-only detectors in two. The largest reversal is on D3. Under LOACO, benign-only and representation scores are close (benign-only minus representation +0.022 [+0.004, +0.047]); under natural novelty, benign-only detectors lead by +0.376 [+0.247, +0.500], and the representation family drops to chance in the mean (0.522 [0.402, 0.647]). Under natural novelty on D1 every family is weak; the best is representation at 0.623 [0.562,

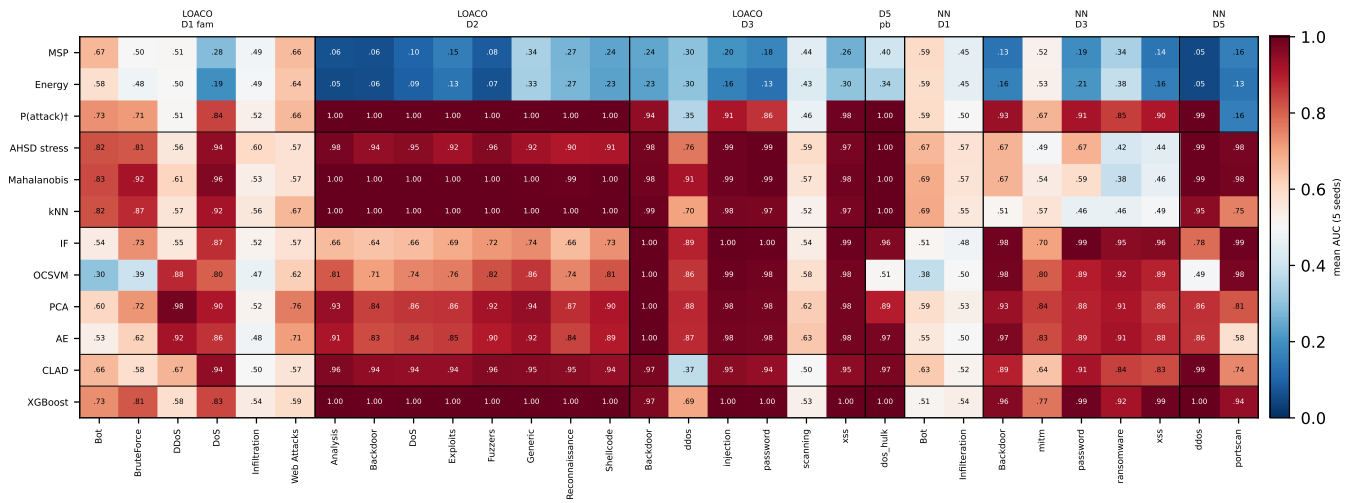


Fig. 5. Mean AUC (five seeds) per detector (rows, grouped by family) and attack class (columns, grouped by protocol and dataset). Red cells are above 0.5 and blue cells below; the leading family changes from one block of columns to the next.

0.692]. **Answer to RQ3 (rankings):** a family selected under LOACO can be the wrong choice under natural novelty, so single-protocol evaluations can crown different winners.

E. C4: A plain supervised classifier is hard to beat

XGBoost exceeds the leading novelty family on LOACO D2 (0.998 vs 0.977), natural novelty D3 (0.923 vs 0.898) and natural novelty D5 (0.968 vs 0.940), and ties it on purged D5, so it matches or beats every dedicated novelty family in four of seven evaluations (Fig. 4). The contrastive zero-day detector CLAD never leads: it trails benign-only detectors on LOACO D3 (-0.123 [-0.277, -0.034]) and natural novelty D3 (-0.075 [-0.130, -0.023]), is below 0.5 on every seed for D3 LOACO ddos (0.372), and beats benign-only detectors on LOACO D2 ($+0.142$ [$+0.119$, $+0.164$]), purged D5 ($+0.139$ [$+0.126$, $+0.148$]) and natural novelty D1 ($+0.069$ [$+0.009$, $+0.146$]). **Answer to RQ3 (supervised reference):** dedicated novelty scores do not reliably beat a plain supervised classifier on held-out or new attacks, contrary to earlier comparisons [21], [17].

F. Confirmatory results: temporal and cross-dataset shift

Two results confirm prior work. Temporal-split effects are dataset-specific (Fig. 6, left), in line with protocol-dependent rankings [15]: on D1 both models lose about 0.27–0.29 AUC, on D2 neither changes by more than 0.014, on D5 both improve, and only on D3 does the neural backbone degrade (0.664, seed std 0.202) while XGBoost holds 0.978. Zero-shot cross-dataset transfer collapses towards chance, as Layeghy and Portmann [29] and Cantone et al. [30] reported: the median of the 108 transfer results (12 detectors $\times$ 9 source–target pairs) is 0.500, with 67% in [0.3, 0.6], 12% below 0.3 and 21% above 0.6; the highest values come from D1$\rightarrow$D3 (CLAD 0.876, XGBoost 0.832). Using the final-epoch instead of the best-validation checkpoint moves representation and CLAD

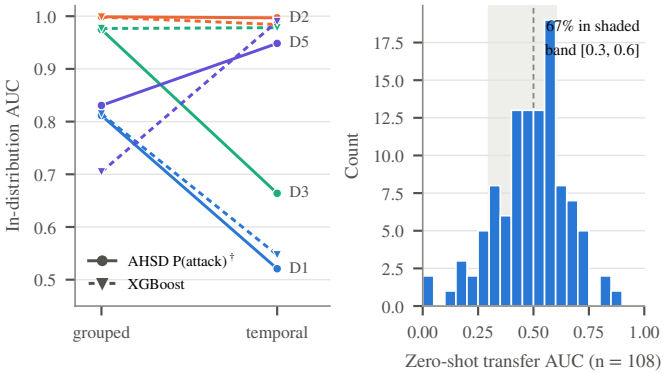


Fig. 6. Left: in-distribution AUC under grouped-random vs temporal splits per dataset (solid: AHSD P(attack); dashed: XGBoost). Right: the 108 zero-shot transfer AUCs (12 detectors $\times$ 9 pairs) cluster around chance, as reported in prior cross-dataset studies.

family means by at most 0.073 AUC and confidence scores by up to 0.205 on D5, whose validation set holds only 26 attack windows; only one benign-only contrast changes sign (natural novelty D1, benign-only minus confidence: -0.025 to $+0.032$).

G. C5: Pre-registered hypotheses that failed

Four confirmatory hypotheses were tested against frozen gates, and all four failed (Table IV). A spectral model of the recurrent filter did not predict per-class detectability (P1). An adaptive equilibrium did not absorb persistent attacks under supervised training; detectability instead peaked at a small adaptation rate (G1). Benign drift with re-anchoring (G2) and benign-only superiority restricted to natural novelty (G3) also failed. G3's supervised family included MSP and energy, which C2 shows are invalid in binary NIDS, so G3 tested a

TABLE IV
PRE-REGISTERED CONFIRMATORY GATES, EVALUATED AS FROZEN. ALL FOUR RETURNED STOP.

Gate	Hypothesis	Frozen rule	Observed
P1	Spectral model predicts per-class detectability	$\rho \geq 0.6$	$\rho = 0.326$
G1	Adaptive equilibrium absorbs persistent attacks	$\rho(\gamma, d') \leq -0.8$	-0.14 / -0.14 / 0.00
G2	Benign drift inverts scores; re-anchoring fixes it	$\geq$5/9 detectors	4, 5, 0 of 9
G3	Benign-only beat supervised only under NN	$B - S \geq 0.15$ / ≤ 0.05	+0.191 / +0.269

flawed contrast; we keep its verdict as recorded and report the corrected family split above.

VII. DISCUSSION

A. Why the protocols disagree

LOACO removes one class but keeps the rest of the training distribution, including benign traffic from the hours of the test set. Natural novelty changes the attack class, the benign traffic and often the hosts and tools at once. Supervised representations are shaped by the attacks they were trained to separate: when the shift is small they place a held-out class far from benign traffic (representation 0.977 on D2 LOACO), but when benign traffic itself drifts, as on D3, the same scores fall to chance and become seed-dependent. Benign-only detectors do not depend on which attacks were seen, which explains their stability but not their accuracy: they lead on D3 natural novelty yet trail by 0.17 on D2 LOACO. XGBoost's strength agrees with evidence that tree ensembles remain hard to beat on tabular data [44].

B. Implications for security management

For an operator, a NIDS is a security service whose assurance depends on evidence gathered before deployment. Our results show that the usual evidence, a single LOACO run scored with MSP or energy, can understate run-to-run variation by up to 17×, raise alarms on the wrong windows, and select the wrong detector family. Because retraining is routine in operation, a detector whose zero-day coverage depends on the random seed is hard to manage: two retrainings of the same model can catch different new attack classes. We therefore recommend the following checklist for detector assurance:

1) Train at least five seeds and report the per-class seed standard deviation, not only the mean; reject detectors whose zero-day coverage depends on the seed.
2) Do not use max-softmax or energy as novelty alarms on binary detectors; use the attack probability or a representation distance, validated under natural novelty.
3) Evaluate under natural temporal novelty wherever timestamps allow, alongside LOACO, and report when LOACO is infeasible.
4) Include a benign-only detector and a gradient-boosted tree as baselines; one of them matched or beat every dedicated novelty method in five of seven evaluations.
5) Group flows into time blocks before windowing, deduplicate windows, and purge neighbouring blocks on short captures.
6) Pre-register acceptance criteria and report the ones that fail.

C. Relation to prior findings

Our results agree with Zoppi et al. that detectors trained without attack labels can handle unknown attacks well [21], but only under some protocols; unlike their study and Mesca et al.'s industrial evaluation [17], we find a supervised XGBoost reference at or above the best novelty family in four of seven evaluations, consistent with protocol-dependent rankings [15]. They do not support a general advantage of training with known attacks [22]: CLAD trails benign-only detectors on both D3 evaluations and beats them on LOACO D2, purged D5 and natural novelty D1. Reported hold-out results near 0.97 [23] are consistent with our LOACO numbers on D2, and our natural-novelty results suggest that they may not hold under temporal shift. Our finding that time-aware evaluation changes conclusions mirrors earlier evidence from malware detection [3] and intrusion detection [25]. The below-chance MSP results reported for multi-class NIDS [13] are consistent with the mechanism we identify in Section VI-C. Day-scheduled captures have been seen as making chronological splits degenerate [16]; we show that the same property yields a usable, and often the only, natural zero-day protocol.

D. Limitations and threats to validity

One neural backbone: all supervised scores come from one recurrent backbone, so Fig. 6 is not a claim about neural networks in general. *Small natural-novelty sets:* D1 and D5 contribute two classes each, and the class bootstrap understates their uncertainty; D3 carries most of the evidence. *Thin validation on D5:* 26 attack windows make checkpoint selection fragile for confidence scores. *Design choices:* the 1-hour grouping, any-attack window label, 200-window threshold and 5:1 test ratio were fixed in advance but are not the only reasonable choices. *Post-hoc addition:* P(attack) is labelled and is not confirmatory evidence. *Reimplementation:* CLAD may differ from its authors' code. *Testbed data:* natural novelty follows each capture's schedule, which is closer to deployment than LOACO but is not live traffic.

VIII. CONCLUSION

Evaluating learned NIDS on natural zero-day traffic, rather than on a single class hold-out run, changes what an operator should believe. Across five datasets, 12 detectors and five seeds, natural novelty made every supervised score 1.7–17× less stable across retraining; max-softmax and energy inverted on binary detectors because confident detection reads as normality; detector-family rankings flipped between protocols; and a plain gradient-boosted tree matched or beat every dedicated novelty family in four of seven evaluations. Four pre-registered hypotheses failed and are reported in full. Future work should add backbones, novel classes and live traffic; code and splits are at [artifact URL].

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